Curriculum Vitæ et Studiorum

Enrico Sartor

Personal and contact information

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Date and place of birth: 14/05/2000, San Vito al Tagliamento (PN), Italy
Citizenship: Italian

1 Education

- **École Doctorale de Mathématique Hadamard (EDMH), Université Paris-Saclay**. *PhD in Applied Mathematics*. Since 01/10/2025
  Host laboratory: Laboratoire des Signaux et Systèmes
  Advisors: Riccardo Bonalli, Laurent Pfeiffer
  Thesis: "Stochastic optimal control under dynamics uncertainty"
- **Università degli Studi di Udine**. *Laurea Magistrale in Matematica* [Master's Degree in Mathematics]. 09/2022–08/10/2024
  Final grade: 110/110 *cum laude* (GPA: 30/30).
  Advisors: Florian Dörfler (ETHZ), Nicolas Lanzetti (ETHZ), Lorenzo Freddi.
  Thesis: "A Pontryagin minimum principle for sparse optimal control in the Wasserstein space".
- **Università degli Studi di Udine**. *Laurea Triennale in Matematica* [Bachelor's Degree in Mathematics]. 09/2019–11/10/2022.
  Final grade: 110/110 *cum laude* (GPA: 29.16/30).
  Advisor: Guglielmo Feltrin.
  Thesis: "Improbability of collisions in the N-body problem".
- **Scuola Superiore Universitaria di Toppo Wassermann**. *Master di secondo livello* [II Level University Master's Degree]. 09/2019-14/03/2024.
  Final grade: 110/110 *cum laude* (GPA: 30/30, *cum laude*). The Scuola Superiore Universitaria di Toppo Wassermann is a higher learning institution based on merit and modelled on the Scuola Normale Superiore di Pisa. The students, selected through a written and oral examination, receive a scholarship and are required to maintain high academic standards and attend additional disciplinary and interdisciplinary courses.

2 Pre-doctoral scientific positions

- **Chair of Dynamics, Control, Machine Learning and Numerics–FAU Erlangen–Nuernberg.** *Vorbereitung eines Promotionvorhabens* [Research training program]. 01/11/2024-31/08/2025.
  Advisor: Enrique Zuazua.

- **ETH Zürich**, **Automatic Control Laboratory**. *NCCR Automation Fellowship Awardee* (visiting Master's student). 01/2024-06/2024.
  Supervisor: Florian Dörfler.

3 Preprints and submitted manuscripts

1. E. Sartor. *A Regular First-Order Calculus for the Wasserstein–Fisher–Rao Geometry.* Preprint, arXiv:2611.09691.
2. E. Sartor. *Optimal control of the coagulation-fragmentation equation.* Preprint, arXiv:2604.13937, submitted for publication, 2026.
3. E. Sartor, F. Dörfler, and N. Lanzetti. *Sparse optimal control in the Wasserstein space.* Preprint, arXiv:2603.00373, submitted for publication, 2026.

4 Talks

1. *Wasserstein-Fisher-Rao differential calculus and optimal control of weighted particle systems*, *PGMO Days 2026*, Fondation Mathématique Jacques Hadamard / EDF Lab Paris-Saclay, Palaiseau (France), 18/11/2026
2. *A Particle Perspective on Partially Observable Stochastic Control*, *Stochastic Optimal Control and Reinforcement Learning*, CIRM, Marseille, France, 09/07/2026.
3. *Optimal Control of the Coagulation-Fragmentation Equation*, *PGMO Days 2025*, Fondation Mathématique Jacques Hadamard / EDF Lab Paris-Saclay, Palaiseau (France), 19/11/2025.

5 Conferences, workshops, and schools

1. *PGMO Days 2026*, Fondation Hadamard, EDF Lab Paris-Saclay (France), 17/11/2026–18/11/2026.
2. *Stochastic Optimal Control and Reinforcement Learning*, CIRM, Marseille, France, 07/07/2026–10/07/2026.
3. *Conference to Honor the Memory of Haïm Brezis*, Sorbonne Université, Paris, France, 22/06/2026–25/06/2026.
4. *Geometry, duality and convexity in new OT problems*, Institut de Mathématique d'Orsay, Orsay (France), 19/11/2025–21/11/2025.
5. *PGMO Days 2025*, Fondation Hadamard, EDF Lab Paris-Saclay (France), 18/11/2025–19/11/2025.
6. *Control and Machine Learning*, IEECI-IGSC, Dubrovnik (Croatia), 30/06/2025–04/07/2025.
7. *Machine Learning and PDEs*, Research Center for Mathematics of Data (Friedrich-Alexander-Universität), Erlangen (Germany), 28/04/2025–30/04/2025
8. *Oberwolfach Seminar: Control and Machine Learning*, Mathematisches Forschungsinstitut Oberwolfach, Oberwolfach (Germany), 24/11/2024–29/11/2024.
9. *18th IFAC Workshop on Time Delay Systems*, International Federation of Automatic Control, Udine (Italy), 24/09/2024–27/09/2024.
10. *PDEs, Control and Deep Learning*, CIME (Centro Internazionale Matematico Estivo), Cetraro (Italy), 22/07/2024–26/07/2024.
11. *Differential equations and dynamics in Alba*, Università degli Studi di Udine, Alba (Italy), 11/09/2023–13/09/2023.

12. *First Italian School on Geometric Deep Learning*, Università degli Studi di Bologna, Pescara (Italy), 25/07/2022–28/07/2022.

6 Teaching activities

1. **Academic year 2026/2027 (fall semester).** CentraleSupélec, Université Paris-Saclay.
   Course: "Convergence, Integration and Probability" (first-year Engineering Programme).
   Role: Teaching assistant (lecturer: Antonio Silveti-Falls).
   Personal teaching load: 15 hours of exercise classes, plus preparation and grading.
   Language of instruction: English.

2. **Academic year 2022/2023 (summer semester).** Università degli Studi di Udine.
   Course: "Analisi Matematica 2" (Bachelor's Degree in Mathematics).
   Role: Tutoring (lecturer: Paolo Baiti).
   Personal teaching load: 30 hours of exercise classes.
   Language of instruction: Italian.

7 Awards, honors, and competitions

1. NCCR Automation Fellowship Awardee, ETH Zürich, 2024.

2. Member of the Scuola Superiore Universitaria di Toppo Wassermann team at the 2023 International Collegiate Programming Contest (ICPC), Southwestern Europe Regional Contest; ranked 55th out of 108 teams. Sorbonne University, Paris, 27–28/01/2024.

3. Admission to the Scuola Superiore Universitaria di Toppo Wassermann, 2019.

4. Silver medal at the national *Individual Italian Mathematical Olympiads* of the Unione Matematica Italiana. Cesenatico (Italy), 03/05/2019.

5. Finalist of the national *Team Italian Mathematical Olympiads* of the Unione Matematica Italiana. Cesenatico (Italy), 04/05/2019.

6. Finalist of the national *Team Italian Mathematical Olympiads* of the Unione Matematica Italiana. Cesenatico (Italy), 05/05/2018.

8 Academic service

1. **Referee for:** *Systems & Control Letters*; *IEEE Control Systems Letters (L-CSS)*.

2. Student representative at Università degli Studi di Udine. 06/2021–08/10/2024.

9 Computer skills

- Programming languages: Python (good knowledge), C, and MATLAB (working knowledge).
- Document preparation system: LaTeX (good knowledge).

Gif-sur-Yvette, 11/11/2026